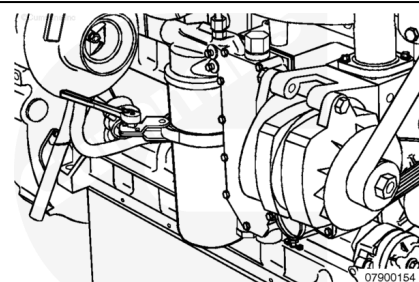


Trainings ▾

Measure

Remove the oil plugs from the lubricating oil cooler housing at the filter inlet and outlet pressure ports and install manual gauges.

The following parts, or equivalent, are available for use:



Part Number	Description
3377244	Compuchek™ fitting (1/8 inch 27 NPT)
3376920	Compuchek™ coupling (1/4 inch pipe thread)
3164491	Electronic pressure adapter for multimeter (1/4-NPT pipe)
3164488 or 3164489	Electronic digital multimeter
Engine Information	
Oil Filter Type	
Miles on Filter	
Oil Type	

Operate the engine at each rpm indicated and record the corresponding pressure values:

	Engine RPM	Oil Pressure Filter Inlet	Oil Filter Pressure Outlet	Inlet - Outlet = Differential Pressure	INSITE™ electronic service tool
Low Idle					
High Idle					

A pressure drop greater than 172 kPa [25 psi], at operating temperature using 15W-40 oil, indicates the filter is plugged.

Identify the causes of a plugged filter. Verify the Cummins Inc. maintenance guidelines are being met.

Use the following procedure to check for possible fluid contamination. Refer to Procedure 007-083 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-083.html>)

Change both the lubricating oil and the lubricating oil filter, if the filter is plugged.

See the following bulletin for additional information about lubricating oil filter plugging.

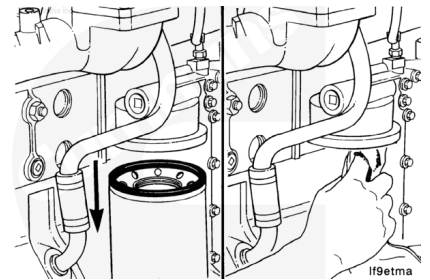
Refer to Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340.
(/qs3/pubsys2/xml/en/bulletin/3810340.html)

Remove

Clean the area around the lubricating oil filter head.

Remove the oil filter with oil filter wrench, Part No. 3375049.

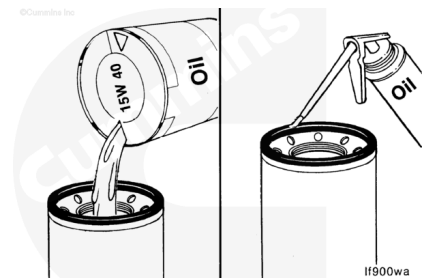
Clean the gasket surface of the filter head.



Install

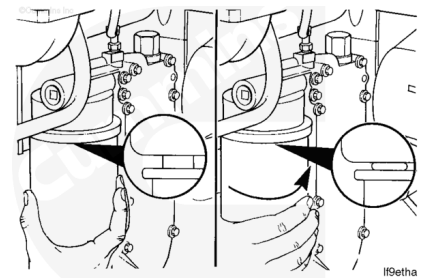
Lubricate the oil filter gasket with clean engine oil.

Fill the oil filter with clean engine oil.

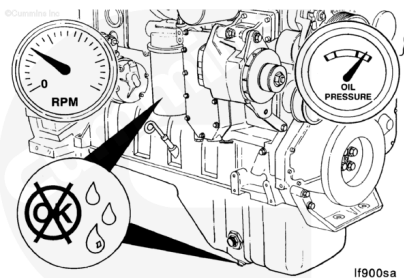


Install the filter on the oil filter head. Tighten the filter until the gasket contacts the filter head surface.

Use oil filter wrench, Part No. 3375049, to tighten the filter an additional $\frac{1}{2}$ to $\frac{3}{4}$ of a turn, or follow the instructions supplied with the oil filter.



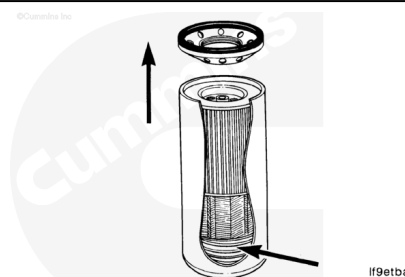
Operate the engine and check for leaks.



Inspect

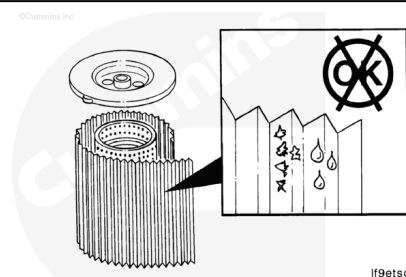
⚠ WARNING ⚠

Carefully cut the full-flow oil filter (upper section of combination filter) open. The filter element spring is under compression and can cause personal injury.



Use the tube cutter, Part No. 3376579, to open the full-flow oil filter (upper section of the bypass filter).

Inspect the filter element for evidence of moisture or metal particles.



Metal	Probable Source
Copper	Bearings and bushings
Chromium	Piston rings
Iron	Cylinder liners
Lead	Bearing overlay material
Aluminum	Piston wear or scuffing